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Multi-source (BFS)

Ques: Rotten Oranges

1 - fresh orange

2 - rotton orange

mat[][] = [[2, 1, 0, 2, 1], [1, 0, 1, 2, 1], [1, 0, 0, 2, 1]]

2	1	0	2	1
1	0	1	2	1
1	0	0	1	1

2	2	2	2	2
2	0	2	2	2
1	0	0	2	1

after 1 hour

2	2	2	2	2
2	0	2	2	2
2	0	0	2	2

after $\boxed{2}$ hours → ans

r-1,c

r,c-1 ← r,0 → r,c+1

r+1,c

(0,0,0) (0,3,0) (1,3,0)

queue < triplet >

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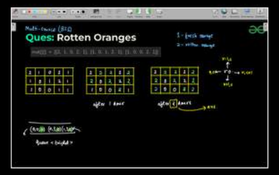
1	1	2
0	0	1
1	0	1

→ not possible to rot 1.



Multi - Source (BFS)

Ques: Number of Enclaves

if bfs is strong, graph is easy

0	0	0	1
0	1	1	0
0	1	1	0
0	0	0	1

0	0	0	1
0	1	0	1
0	0	1	0
0	1	0	1
0	0	0	0

⇒

0	0	0	-1
-1	0	-1	-1
0	0	-1	0
0	1	0	-1
0	0	0	0

ans = 4

1) if there is a 1 on boundary then it can go out

2) all 1's which can reach a boundary 1, they can go out

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Ques:

Distance of nearest cell having 1

1	0	0
0	1	0
0	0	1

arr

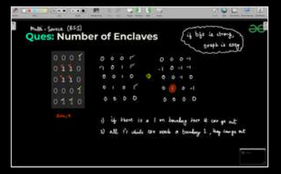
→

0	1	2
1	0	1
2	1	0

ans

1 need dist to nearest 1

Logic - q-insert all ones (row, col, 0)

dist

dist = ans[row][col]